

Human-Computer Interaction

Experimental Design

+ Measurement Basics

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Today's Agenda

- Part 1: *Experimental research basics*
- Part 2: *A step-by-step design guide*
- Part 3: *Measurement basics*
- Hands-on: *reverse-engineer an experiment*

Due today, 9:30 am

- **Qualitative Field-Method Plan** (*individual*)
- Your team's **3-line project status**: done / blocker / next

Reading for today

Lazar, Feng & Hochheiser (2017), *Research Methods in Human-Computer Interaction*

- **Ch. 2:** Experimental research
- **Ch. 3:** Experimental design

Looking ahead to the Oct 16 crit: Ch. 5 (Surveys) and Ch. 13 (Measuring the human).

Part 1: Experimental Research Basics

What is a hypothesis?

A testable statement of the predicted relationship between at least two variables, answering a *research question*.

Question: How does information on a caller's context affect whether the receiver picks up?

Hypothesis: Receivers will be more likely to pick up when they have information on the caller's context than when they do not.

Two features: **falsifiable** and **testable**.¹

¹Popper, K. (1959). *The Logic of Scientific Discovery*. Hutchinson.

Good hypotheses are

1. **Testable**: you can manipulate IVs and measure DVs
2. **Falsifiable**: the data could prove them wrong
3. **Parsimonious**: the simplest form of the prediction
4. **Precise**: operationalizable
5. **Useful**: informs existing or new theory

Five types of variable²

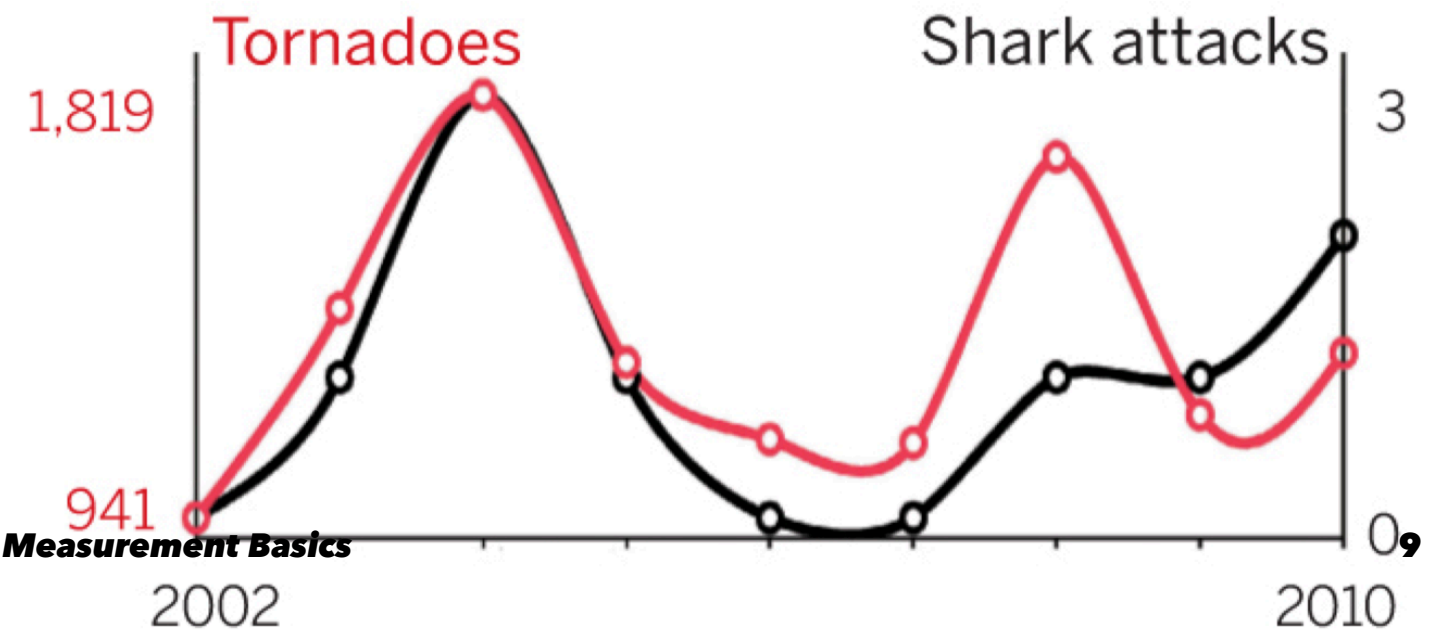
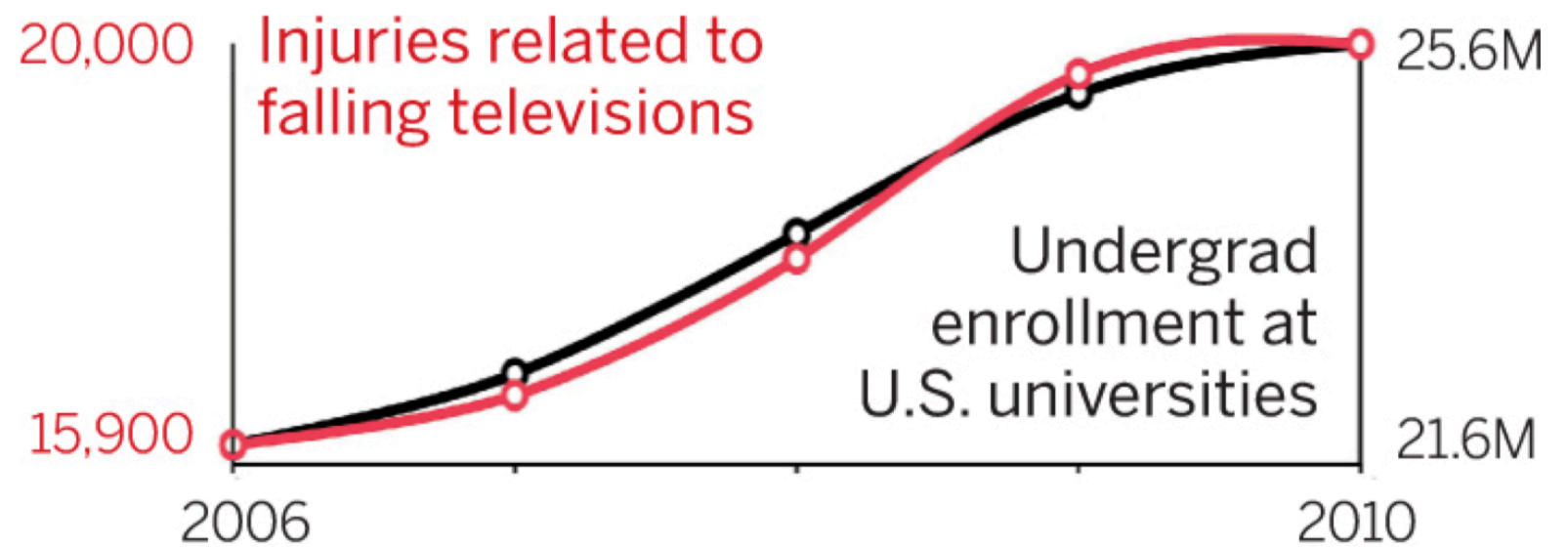
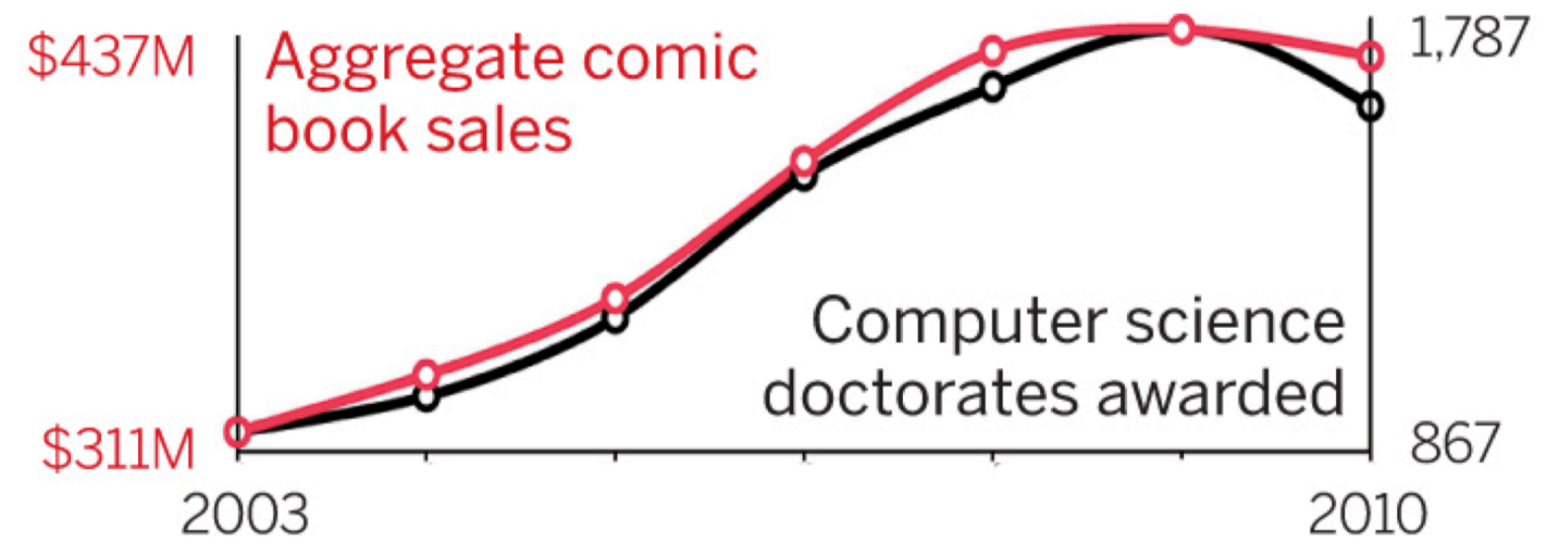
1. **Independent**: what is manipulated
2. **Dependent**: what is measured
3. **Control**: what is held constant
4. **Random**: what is allowed to vary randomly
5. **Confounding**: what correlates with IV/DV

² McGrath, J. E. (1995). Methodology matters. In *Readings in HCI* (pp. 152-169). Morgan Kaufmann.

What relationships are we after?

- **Causal:** one variable (DV) depends on and is affected by another (IV)
- **Correlational:** two variables move together, possibly driven by a third

*Correlation $\neq$ causation.*³



³ Charts: Vigen, T. (2015). *Spurious Correlations*. Hachette.

Three questions → three designs

Do we *know* all variables? Can we *control* the IV? Can we *measure* all variables?

- **Yes / Yes / Yes → true experiment** (manipulate IV, measure DV, cause-effect claim)
- **Yes / No / Yes → quasi-experiment** (split on a characteristic; no random assignment)
- **No / any / Yes → correlational study** (relationships among measured variables)

Factors, levels, assignment

IVs = **factors**; their values = **levels** (conditions). We compare levels' effects on the **response**.

Within-participants

Everyone sees every level (in order)

+ power · – transfer effects

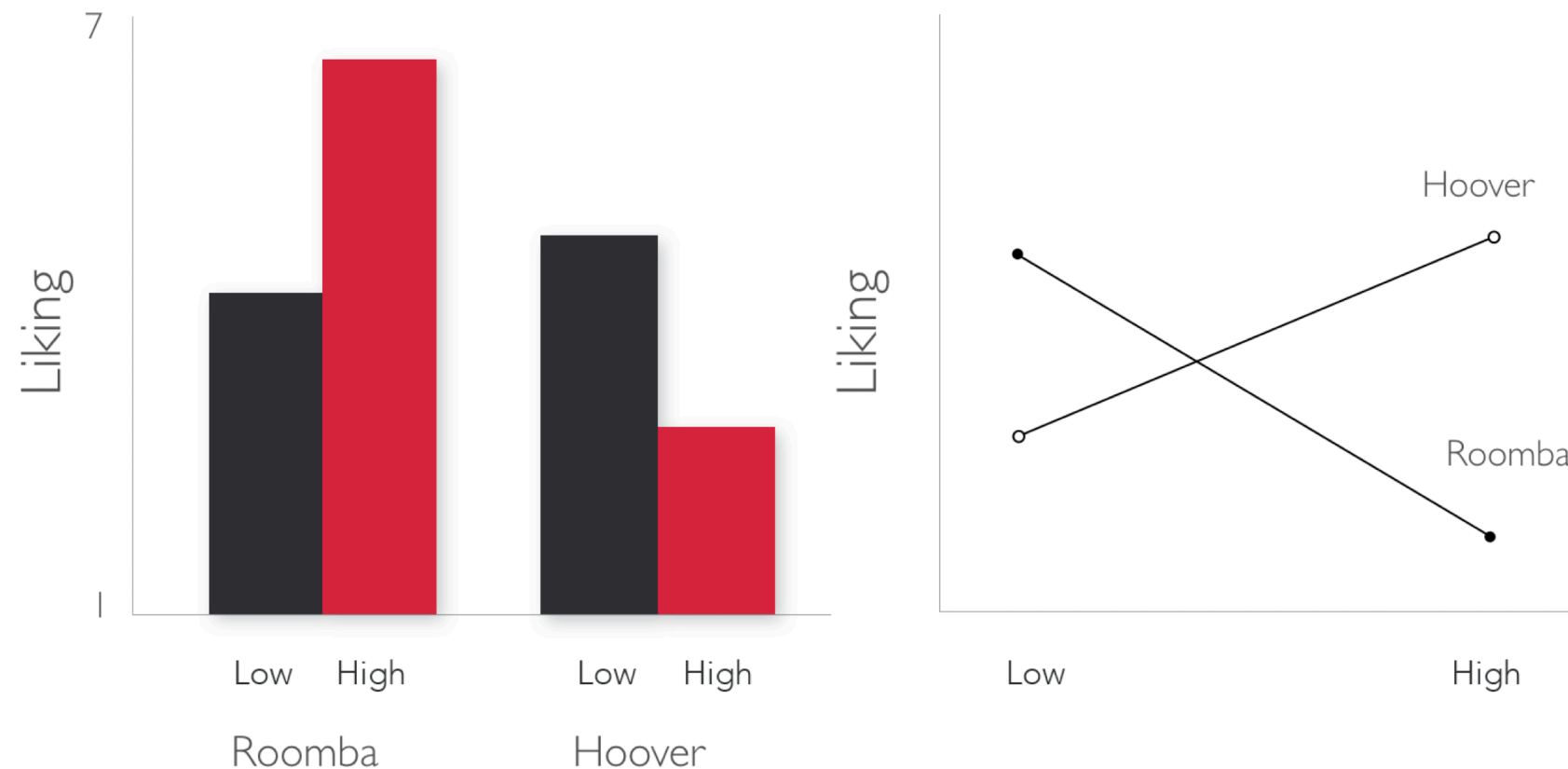
Between-participants

Each subgroup sees one level
+ no transfer · – needs larger n

Multiple factors → factorial designs

Manipulate two or more IVs at once, e.g. gaming (low/high) x tech-savviness (low/high).

- **Main effect:** effect of a single factor on the response
- **Interaction effect:** the effect of one factor depends on the level of another
- More efficient, but more demanding (larger n)



Controlling for bias

Random sampling + assignment protect validity and generalizability.

- **Counterbalancing**: vary level order across participants to defuse *transfer effects* (learning, fatigue)
- **Blocking / Latin square**: structured counterbalancing
- **Asymmetrical transfer** → prefer a *between-participants* design
- **Blinding**: single- / double-blind keep stakeholders naive to condition

Part 2: Step-by-Step Design Guide

The seven steps

1. **Formulate** the research question
2. **Identify** variables
3. **Generate** hypotheses
4. **Determine** the experimental design
5. **Develop** task & procedure
6. **Determine** manipulations & measurements
7. **Identify** participants

Step 1: Research question

The central issue the study resolves. It should be *specific*, name the *variables*, and state the *conditions* Z.

FINER: **F**easible · **I**nteresting · **N**ovel · **E**thical · **R**elevant⁴

⁴Farrugia et al. (2010). Research questions, hypotheses, and objectives.

Steps 2-3: Variables → hypotheses

- **Identify variables:** IVs (factors + levels), DVs, random factors
- **Hypothesize** in the canonical form:

RQ: What are the effects of X on Y under conditions Z ?

H: Y will be higher/lower when X is X_i than when X is X_j , under Z .

Sources: exploratory results · related theory · reasoned face validity;
always justified.

Step 4: Experimental design

How participants are allocated to conditions:

- **Within-participants** (repeated measures): power, fewer n ; transfer/demand risk
- **Between-participants** (independent measures): no transfer; more n , more variance
- **Mixed-model** (split-plot): some factors within, some between

Prefer factorial (2^k) designs where feasible.

Step 5: Task & procedure

Task: a reasonable representation of the real-world context Z ; relevant, controllable, motivating.

- **Distractor tasks** add cognitive or perceptual load
- **Deceptive tasks** use a cover story to reduce demand characteristics

Procedure: a replicable step-by-step of instructions, roles, equipment, and timeline (consent → measures → compensation).

Steps 6-7: Manipulations & participants

- **Manipulations & measurements:** IVs can be *manipulated* or *measured*; control levels give a *baseline* or *gold standard*
- **Participants:** a representative, sufficiently large, balanced sample of the target population (Z)

Everything not manipulated must be measured, which is Part 3.

Part 3: Measurement Basics

What do we measure when we measure?

Measurement: assigning a number to a characteristic so it can be compared.

- **Quantitative:** the degree of an attribute (marathon time, SAT score)
- **Qualitative:** a description ("the first customer was a teenage boy")

Four types of variable

From least to most precise:

- **Nominal**: categories (novice / expert)
- **Ordinal**: ranked, unequal gaps (very satisfied → very unsatisfied)
- **Interval**: equal gaps, no true zero (1–7 rating)
- **Ratio**: equal gaps, real zero (time, weight)

Four kinds of measure

1. **Objective**: against a standard (test score, typing speed)
2. **Behavioral**: actions and behaviors (eye contact with a robot)
3. **Subjective**: self-report (preferences, personality)
4. **Physiological**: from the body (GSR, EEG, EMG, fMRI)

What makes a measure good?

1. **Validity**: are we measuring what we intend? (*face, construct, criterion, content, ecological*)
2. **Reliability**: is the measure consistent across repeats? ($\alpha \geq .70$ desirable)
3. **Quality**: is error controlled? (*random = noise; systematic = bias*)

Full treatment (scales, α , power) at the Oct 16 Method-Design Crit.

Hands-on: reverse-engineer an experiment (30 min)

Pods	Paper	
1-5	A · Jakesch et al. (2023)	Co-writing with opinionated LMs
6-9	B · Dhillon et al. (2024)	Scaffolding levels in co-writing
10-13	C · Buçinca et al. (2021)	To trust or to think

Abstract + intro (5 min) → method (15 min) → debrief (10 min).
Nothing to submit.

Grab the worksheet

- One copy per pod: File → Make a copy
- Nothing to submit today
- Each item maps to a step of your **Step-by-Step Experimental Design**

da.gd/ZA9eH



Debrief: three papers, three designs

	A • Jakesch	B • Dhillon	C • Buçinca
Assignment	between, random	within, Latin square	conditions randomized by block
Hypotheses	none stated	research question	explicit H1a, H1b, H2
People	1,506 online (Prolific)	131, live on Zoom	199 (Mechanical Turk)

What's Next?

- **Monday: Accessibility, Aging & AI**
 - **Read:** Wobbrock et al. (2011) · Glazko et al. (2023)
 - Post your provocation slide
- **Due Fri Oct 16, 9:30 am:** Step-by-Step Experimental Design (*individual*)
- **Oct 16:** Method-Design Crit; bring the design